Bias-Variance decomposition expanded with details

(Week 4, slide 6 of "W4-L2.pdf")

Goal

- Show for a scalar estimator $\hat{\theta}$ and true parameter θ^*:

$$\mathbb{E}\left[(\hat{\theta} - \theta^*)^2\right] = \mathsf{Var}[\hat{\theta}] + \mathsf{Bias}[\hat{\theta}]^2.$$

- Expectation $\mathbb{E}[\cdot]$ is taken over the distribution of training sets (i.e. randomness in $\hat{\theta}$).

Start: expand the square

Compute the expected squared error by expanding the square inside the expectation:

$$\begin{aligned}\mathbb{E}\big[(\hat{\theta} - \theta^*)^2\big] &= \mathbb{E}\big[\hat{\theta}^2 - 2\theta^*\hat{\theta} + (\theta^*)^2\big] \\ &= \mathbb{E}[\hat{\theta}^2] - 2\theta^*\,\mathbb{E}[\hat{\theta}] + (\theta^*)^2.\end{aligned}$$

Note: linearity of expectation lets us pull constants (like θ^*) out of $\mathbb{E}[\cdot]$.

Add and subtract $\mathbb{E}[\hat{\theta}]^2$

To separate variance and bias, add *and* subtract $\mathbb{E}[\hat{\theta}]^2$:

$$\begin{aligned}\mathbb{E}\left[(\hat{\theta}-\theta^*)^2\right] &= \mathbb{E}[\hat{\theta}^2] - 2\theta^*\mathbb{E}[\hat{\theta}] + (\theta^*)^2 + \mathbb{E}[\hat{\theta}]^2 - \mathbb{E}[\hat{\theta}]^2 \\ &= \left(\mathbb{E}[\hat{\theta}^2] - \mathbb{E}[\hat{\theta}]^2\right) + \left(\mathbb{E}[\hat{\theta}]^2 - 2\theta^*\mathbb{E}[\hat{\theta}] + (\theta^*)^2\right).\end{aligned}$$

We grouped terms so the first group looks like a variance and the second group is a perfect square in $\mathbb{E}[\hat{\theta}]$ and θ^*.

Recognize variance and bias^2 (part 1)

The first bracket:

$$\mathbb{E}[\hat{\theta}^2] - \mathbb{E}[\hat{\theta}]^2 = \mathsf{Var}[\hat{\theta}],$$

Check: By definition, the variance of a random variable $\hat{\theta}$ is

$$\mathsf{Var}[\hat{\theta}] := \mathbb{E}\left[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2\right].$$

Expand the square inside the expectation:

$$\mathsf{Var}[\hat{\theta}] = \mathbb{E}\left[\hat{\theta}^2 - 2\hat{\theta}\,\mathbb{E}[\hat{\theta}] + (\mathbb{E}[\hat{\theta}])^2\right].$$

Use the property that $\mathbb{E}[\cdot]$ is linear, and note that $\mathbb{E}[\hat{\theta}]$ is a constant:

$$\mathsf{Var}[\hat{\theta}] = \mathbb{E}[\hat{\theta}^2] - 2\,\mathbb{E}[\hat{\theta}]\,\mathbb{E}[\hat{\theta}] + (\mathbb{E}[\hat{\theta}])^2.$$

Simplify terms:

$$-2\mathbb{E}[\hat{\theta}]^2 + (\mathbb{E}[\hat{\theta}])^2 = -\mathbb{E}[\hat{\theta}]^2.$$

Substitute back:

$$\mathsf{Var}[\hat{\theta}] = \mathbb{E}[\hat{\theta}^2] - (\mathbb{E}[\hat{\theta}])^2.$$

Numerical Example

$$\mathbb{E}[\hat{\theta}^2] - \mathbb{E}[\hat{\theta}]^2 = \mathsf{Var}[\hat{\theta}].$$

Suppose $\hat{\theta}$ takes values 0 and 2 each with probability $1/2$.

$$\begin{aligned} \mathbb{E}[\hat{\theta}] &= \ldots\ldots\ldots\ldots\ldots\ldots = 1, \\ \mathbb{E}[\hat{\theta}^2] &= \ldots\ldots\ldots\ldots\ldots\ldots = (0^2 + 2^2)/2 = 2, \\ \mathsf{Var}[\hat{\theta}] &= 2 - 1^2 = 1. \end{aligned}$$

This matches the variance computed directly: $\frac{1}{2}[(0-1)^2 + (2-1)^2] = 1$.

Recognize variance and bias^2 (part 2)

The second bracket:

$$\mathbb{E}[\hat{\theta}]^2 - 2\theta^*\mathbb{E}[\hat{\theta}] + (\theta^*)^2 = (\mathbb{E}[\hat{\theta}] - \theta^*)^2.$$

Define the bias of $\hat{\theta}$ as

$$\mathsf{Bias}[\hat{\theta}] := \mathbb{E}[\hat{\theta}] - \theta^*.$$

Therefore the second group equals $\mathsf{Bias}[\hat{\theta}]^2$.

Conclusion

Putting the two pieces together:

$$\mathbb{E}\big[(\hat{\theta} - \theta^*)^2\big] = \text{Var}[\hat{\theta}] + \text{Bias}[\hat{\theta}]^2$$

- This is the canonical *bias–variance decomposition* for scalar estimators.
- It shows the expected squared error decomposes into the variance of the estimator and the square of its bias.

Optional remarks

- If $\hat{\theta}$ is unbiased (i.e. $\mathbb{E}[\hat{\theta}] = \theta^*$), the bias term vanishes and
$$\mathbb{E}\big[(\hat{\theta} - \theta^*)^2\big] = \mathsf{Var}[\hat{\theta}].$$

- For vector-valued parameters the decomposition generalizes using the mean squared error matrix:
$$\mathbb{E}\big[(\hat{\theta} - \theta^*)(\hat{\theta} - \theta^*)^\top\big] = \mathsf{Cov}(\hat{\theta}) + \mathsf{Bias}(\hat{\theta})\,\mathsf{Bias}(\hat{\theta})^\top.$$

- Assumes expectations exist (finite second moments).